Arunabha Majumder

Mechanical Engineer | Robot Joints, Actuators and Linkages | Prototyping and Testing

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SUMMARY

I design robot joints and actuators and then build and test them myself. For my PhD I designed a shoulder exoskeleton that works next to the human arm, using a spring and a small 6 Nm motor that share the load. It cut energy use by more than 25% and won a Best Paper Award in 2026. I work in SolidWorks (CSWA), prototype with 3D printing, and sit with the electronics and software people until the whole thing runs.

KEY RESULTS

- **>25%** less energy from a new spring and motor actuator for a shoulder joint
- **>15%** less muscle effort for the user, tested on 12 people
- **<10%** force error on a 5-bar linkage robot, checked against a force sensor

EXPERIENCE

Research Engineer, Exoskeleton Joints and Actuators (PhD project VIEXO) | 2023 - Present
Aalborg University, Aalborg, Denmark

- Cut energy use by more than 25% (Best Paper Award, IFToMM ISRM 2026) by designing the joint, frame and actuator of a shoulder exoskeleton in SolidWorks, where a parallel spring carries part of the load and a 6 Nm motor handles the rest.
- Got force estimates within 10% RMS of a reference force sensor by working out the linkage kinematics (Jacobian) of a 5-bar planar parallel robot driven by variable-stiffness actuators.
- Went through several versions of each part in weeks rather than months by 3D printing them, fitting them on the rig and fixing what broke or flexed too much.
- Lowered user muscle effort by more than 15%, measured with EMG on 12 participants, after fitting the motor, encoder and force sensors into the frame and working with two colleagues on the controller (my part: equations such as the Fourier fit, and the GUI behaviour).
- Built my own test rigs to check torque, motion and durability of each version, and kept the drawings, parts lists and test notes so others could rebuild the setup. Supervised Bachelor's students (2023 - 2025).

MTech Researcher, Pneumatic Actuators (Master's thesis) | 2020 - 2022
CSIR-Central Mechanical Engineering Research Institute (CMERI), Durgapur, India

- Brought position error down to 0.3-0.78%, better than a standard PID, by working on the outer PID loop of a neural-network gain-scheduled controller in MATLAB/Simulink. I collected the experimental data, fed it to the lab's existing neural network and used its output to set the PID gains (IEEE CONECCT 2022).
- Fabricated and assembled the muscles to an existing lab design and tested them on the lab's setup (valves, pressure sensors, DAQ) to map load against stroke.

Bachelor's Capstone, Upper-Limb Rehabilitation Exoskeleton | 2018 - 2019
Siddaganga Institute of Technology, Tumakuru, India

- Designed it in CAD, machined the parts, assembled it and tested it with users. Best Major Project Award and an INR 100,000 TATA Technologies grant.
- Also designed, machined and put together a Go-Kart (2017) and a human-electric hybrid vehicle for Efficycle (2018).

SKILLS

- **Mechanical design:** SolidWorks (CSWA), parts, assemblies and drawings, joints, linkages, actuator layouts
- **Engineering basics:** Statics, dynamics, kinematics and Jacobians, choosing parts for strength and weight
- **Building:** 3D printing (Bambu Lab, Ultimaker Cura); CNC and manual machining and assembly (Bachelor's)
- **Integration and test:** Motors, encoders, force/torque sensors, pressure sensors, DAQ, Arduino; Python, MATLAB/Simulink
- **Languages:** English (C1, fluent), Bengali (native), Hindi (conversational)

EDUCATION

PhD, Mechatronics and Robotics (in progress) | 2023 - Present
Aalborg University, Department of Materials and Production, Denmark

MTech, Mechatronics | 2020 - 2022
AcSIR - CSIR-CMERI, Durgapur, India

BTech, Mechanical Engineering | 2015 - 2019
Siddaganga Institute of Technology (VTU), Tumakuru, India

AWARDS AND PUBLICATIONS

- Best Research Paper Award, IFToMM ISRM 2026; Certified SolidWorks Associate (CSWA)
- Majumder, A., et al. "A Hybrid Actuated Shoulder Exoskeleton for Energy-Efficient Upper Arm Support." ISRM 2026, Springer.
- Majumder, A., et al. "Neural Network-Based Gain Scheduled Position Control of a Pneumatic Artificial Muscle." IEEE CONECCT, 2022.

REFERENCES

Available upon request.